

Question Paper

Exam Date & Time: 01-Jan-2026 (09:30 AM - 12:30 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MANIPAL INSTITUTE OF TECHNOLOGY - BENGALURU
THIRD SEMESTER B.TECH. DEGREE MAKEUP EXAMINATIONS - JANUARY 2026
COMPUTER ORGANIZATION and ARCHITECTURE [CSE_2121]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) Consider that a list of 32-bit positive integers are stored in the memory from the location named LIST. The word at address ELE is to hold the value of the smallest number after it has been found in this list. Assuming, the number of entries in the list is present at the memory location N, Write a RISC-style program to find the smallest number in the list. (4)
- A)
- Course Outcome: "CO1" / Cognitive Level: "Analysis"
- B) Perform floating point multiplication of the following numbers and represent the result in IEEE 32 bit format. (4)
- $X = 3.344 * 10^1$ and $Y = 8.877 * 10^{-2}$
- Course Outcome: "CO1" / Cognitive Level: "Application"
- C) Explain how the Restoring Division Algorithm is used to divide the numbers when the dividend = 7 and the divisor = 4. (2)
- Course Outcome: "CO1" / Cognitive Level: "Understand"
- 2) Design a Register Transfer Level (RTL) description to compute the factorial of a given number using the input bus (INBUS), registers, and an adder within the Processing Unit (PU). Clearly identify the control points along with the corresponding actions performed by the processing unit at each stage. Additionally, draw an efficient state diagram indicating the control points and transitions required to implement the factorial computation. (4)
- A)
- Course Outcome: "CO2" / Cognitive Level: "Evaluate"
- B) Evaluate with Non-Restoring division algorithm to perform the following division 14/3 (4)
- Course Outcome: "CO2" / Cognitive Level: "Evaluate"
- C) Show flowchart to illustrate the step-by-step process of restoring division method in detail (2)
- Course Outcome: "CO2" / Cognitive Level: "Understand"
- 3) Evaluate the efficiency and address structure of a 2-way set-associative cache having a total capacity of 64 KB and a block size of 128 bytes, when the main memory size is 256 MB. Find the number of bits required for the TAG, SET, and WORD (block offset) fields and how the chosen cache parameters affect address mapping efficiency, hit rate, and hardware complexity in the memory hierarchy. (4)
- A)

Course Outcome: "CO3" / Cognitive Level: "Evaluate"

- B) A computer system has a byte-addressable memory and a data cache capable of storing eight words, where each cache block contains one word (one byte). The cache is initially empty. During program execution, the processor reads data sequentially from the following memory addresses (in decimal): 1024, 1025, 1032, 1037, 1024, 1040, 1032, 1025, 1039, 1025, 1037, 1042, 1044, 1037, 1041, 1043 (4)
- Using the given information:
1. Apply the LRU replacement algorithm to determine the cache contents after each memory access.
 2. Compute the cache hit rate for this reference sequence.

Course Outcome: "CO3" / Cognitive Level: "Application"

- C) Consider a long sequence of accesses to a disk with an average seek time of 4 ms and an average rotational delay of 2 ms. The average size of a block being accessed is 4K bytes. The data transfer rate from the disk is 34 Mbytes/sec. Assuming that the data blocks are randomly located on the disk, estimate the average percentage of the total time occupied by seek operations and rotational delays. (2)

Course Outcome: "CO3" / Cognitive Level: "Understand"

- 4) Consider a block set-associative cache consisting of 64 blocks that are divided into 4 block sets. The main memory contains 4096 blocks, where each block has 128 words, and each word is 16 bits long. (4)
- A)
1. Illustrate the organization of the cache with a neat and properly labelled diagram.
 2. Determine:
 - i) The total number of bits in the main memory.
 - ii) The number of bits required in each of the TAG, SET, and WORD fields of the address.

Course Outcome: "CO4" / Cognitive Level: "Analysis"

- B) Evaluate the design requirements and organization strategy for constructing a $2G \times 16$ memory module using $256K \times 8$ static memory chips. In your evaluation, justify: The number of chips required to achieve the desired memory capacity. The memory organization and interconnection scheme between chips. The advantages and limitations of your proposed design in terms of scalability, performance, and hardware complexity. (4)

Course Outcome: "CO4" / Cognitive Level: "Evaluate"

- C) Contrast between bandwidth and effective throughput of an interconnection network. Justify the need for arbitration when bus interconnection network is employed in a shared-memory multiprocessing architecture. (2)

Course Outcome: "CO4" / Cognitive Level: "Understand"

- 5) Define hit rate and miss penalty. Evaluate the total access time if miss rate is 0.12 miss penalty is 0.015ms and cache access time is 10microseconds. (4)

A)

Course Outcome: "CO5" / Cognitive Level: "Evaluate"

- B) Analyze the five-stage pipelining architecture by interpreting its diagram and examining how each interstage buffer contributes to instruction flow, hazard reduction, and overall pipeline efficiency. (4)

Course Outcome: "CO1" / Cognitive Level: "Analysis"

- C) What are the critical stages involved in the instruction processing of a pipelined processor, and how do the stages contribute to execution efficiency? (2)

Course Outcome: "CO4" / Cognitive Level: "Understand"

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Event Name	Subject
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058-MIT BLR-MAKEUP EXAM-2025-2026 CSE_2121:COMPUTER ORGANIZATION and ARCHITECTURE

#	Course Outcome	Description	Marks
1	1	CO 1	14
2	2	CO 2	10
3	3	CO 3	10
4	4	CO 4	12
5	5	CO 5	4
Total			50

* **NA** - CO Not Selected

Event Name	Subject
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058-MIT BLR-MAKEUP EXAM-2025-2026 CSE_2121:COMPUTER ORGANIZATION and ARCHITECTURE

#	Level	No of Questions	Marks (%)
1	Remember	0	0(0%)
2	Understand	5	10(20%)
3	Application	2	8(16%)
4	Analysis	3	12(24%)
5	Create	5	20(40%)
6	Evaluate	0	0(0%)
Total		15	50 (100%)